

SATHWIK JANNU
NETWORK ENGINEER
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Profile Summary

- Accomplished Data Engineer and Python Developer with 4 years of hands-on experience designing, implementing, and optimizing data architectures and scalable data pipelines.
- Adept at leveraging big data technologies such as Apache Spark, Hive, Impala, and Sqoop for data processing, migration, and transformation at scale. Skilled in building robust ETL/ELT pipelines and automating workflows with Azure Data Factory, Oozie, and Autosys to streamline data operations and ensure seamless data integration across platforms.
- Deeply experienced in working with cloud environments, including AWS and Microsoft Azure, where I have led the design and deployment of cloud-native solutions, utilizing AWS EMR, S3, Lambda, Azure SQL Database, and Azure Data Lake for optimal data storage, processing, and management.
- Proficient in developing complex data models and data warehousing solutions, with expertise in PostgreSQL, MongoDB, MySQL, MSSQL, and Oracle to manage and structure large, high-volume datasets.
- Expert in machine learning and data science workflows, utilizing Python, NumPy, Pandas, Scikit-learn, and Matplotlib to drive predictive analytics and automate business processes.
- Applied advanced statistical learning techniques (e.g., Logistic Regression, Random Forest) to solve complex problems in fraud detection, financial crime prevention, and monitoring of politically exposed persons.
- Proficient in full-stack development with Java, Python, R, Scala, JavaScript, and C++, with substantial experience in creating REST APIs, developing microservices, and designing web applications using Spring Boot, Flask, Django, AngularJS, and Node.js. Experienced in containerization with Docker, and orchestrating complex systems with Kubernetes for high-availability and scalable deployment in cloud environments.
- Strong advocate of CI/CD principles, leveraging Jenkins, Git, and SVN for version control and continuous deployment automation.
- A proactive problem solver, skilled in automating system operations through Shell Scripting and optimizing data pipelines for performance. With a solid foundation in Agile methodologies, Worked in cross-functional teams to drive technical solutions that are reliable, efficient, and aligned with business goals.

SKILLS & INTERESTS

Programming Languages:	Java, Python, R, HTML, XML, Shell, C, C++, Spark, Scala, JavaScript
Web Technologies:	JavaScript, REST API, Spring, Springboot, Microservices, XML, HTML5, CSS3, Bootstrap.
Databases:	MySQL, MS SQL, MongoDB, MSSQL.
Tools:	Docker, Jenkins, Kubernetes.
Machine Learning:	NumPy, Pandas, Scikit-learn, Matplotlib, Statistical Learning Methods.
Frameworks & IDEs:	Agile Methodologies, Visual Studio Code, Jupyter Notebook, Google Colab, Django, Flask, Angular JS, Node JS, Pyramid.
Cloud:	AWS, Microsoft Azure, Cloudera.
Miscellaneous:	Shell Scripting, Servlets, JSP, CI/CD, Windows and Linux.

Work Experience

Data Engineer
Ameriprise financial, Charlotte,NC

May 2024 - Present

- Worked with Business Analyst and helped representing the business domain details, prepared documentation.
 - Helped the Business Insights team with statistical predictions, business intelligence and data science efforts.
 - Teamed up with Architects to design Spark model for the existing Mapreduce model Created Hive/Impala tables and created Sqoop jobs to import the data from Oracle to HDFS and scheduled them in Autosys to run every day.
 - Developed scripts, UDF's using both Spark SQL and Spark Core in Python for aggregative operations Experienced in implementing Spark RDD/Data Frame transformations, actions to implement business analysis and Worked with Spark accumulators and broadcast variables.
 - Analyzed the SQL scripts and Designed the Solution to Implement Using PySpark.
 - Efficiently joined raw data with the reference data using Pig scripting. Used various file formats like Parquet, Avro, ORC and compression techniques like Snappy, LZO and GZip for efficient management of cluster resources.
 - Developed Oozie workflows to automate the jobs and Used SVN for version control on the project.
 - Cloudera Manager was used to monitor the health of Jobs which are running on the cluster
 - **Environment:** Hadoop,Hive, Sqoop, Spark, Oozie, Impala, SQL, Tableau, Python and Autosys
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Data Engineer

All State, Charlotte, NC

August 2023 - May 2024

- Provided weekly data insights on the politically exposed persons, sanction lists, and monitoring their activity to help prevent financial fraud and identify potential criminals in the industry using media articles as the source.
- Created and deployed data dashboards on docker to visualize the inquiries querying in SQL to retrieve data as per client inquiries providing daily reports using production data and experimental analysis using development data.
- Performed Data Wrangling with MongoDB documents having more than 66 Million records to extract features that contained metadata including culture, foreign language name, geo-distance, and politically exposed persons.
- Investigated between open source Metabase and subscription based ZoomData to perform load tests and generated reports based on performance and integrated with the EMR (AWS) cluster.
- Applied Logistic Regression after one hot encoding the extracted features matched with the query submitted by client to predict the alert generated manually by analysts in order to automate the analyst review system.

Skills: Python, Spark, Docker, MongoDB, Linux, AWS, Matplotlib, Scikit-learn, Node JS, Java

Jr. Data Engineer

Razor sight Pvt Ltd

August 2021 - August 2022

- Development and deployment of ETL/ELT migration pipelines for full and incremental loading from onprem server.

- Created lift-and-shift pipelines from SQL Server to Azure Managed Postgres Instance.
- Database Denormalization ETL/ELT with PostgreSQL and Azure Data Factory.
- Worked with stakeholders to create metadata workbooks for mapping denormalized tables.
- Created multi-layer aggregation process for denormalized schema.
- Created pipelines with ADF to generate and store metadata for denormalized schema.
- Created custom methods for catching delta data for updating the denormalized schema.
- Report Conversion and Delivery with PostgreSQL and Azure Data Factory and Logic Apps.
- Translated existing reports from MSSQL to PostgreSQL language.
- Created publishing pipelines for access and delivery of reports on schedule.
- Created an Azure Logic App for generating SAS keys for reports in Azure Blob Storage.
- Developed Logging Method for Data Acquisition.
- Table design for logging multiple acquisition methods in PostgreSQL.
- Created generic pipeline in ADF to log variety of actions from a variety of pipelines.
- Development of a currency conversion solution.
- Serverless Azure Function development for data acquisition in python.
- PostgreSQL table design and management.
- Sisense ElastiCube development for integration into the suite of dashboards.
- Lead Support of an Azure-based Flask Microservice for Custom “Drag and Drop” Reporting Tool

Environment:Azure,Microservice,PostgreSQL,MSSQL,Azure data factory,ETL/ELT,Flask

Jr. Python Developer

XLNCI BIZ Solutions, Hyderabad, India

Jan 2021 – July 2021

- Developed views and templates with Python and Django's view controller and templating language to create a user-friendly website interface.
- Used Python and Django creating graphics, XML processing, data exchange and business logic implementation. Utilized Python Libraries like Boto3 for AWS.
- Managed Container using Docker by writing Docker les and set up the automated build on Docker HUB and installed and congured Kubernetes.
- Generated graphical reports using Python package NumPy and used advanced features like pickle/unpickle in Python for sharing the information across the applications to deploy into AWS with tools/platforms such as JIRA, Jenkins, Docker, and Elastic search.
- Implemented Jenkins in a Docker container with EC2 slaves in Amazon AWS cloud conguration. Used Beautiful Soup, Matplotlib and Pandas Libraries to create access weather data and perform data analysis and visualization.
- Used JSON based REST web services and Amazon web services (AWS) for smooth operation of application.
- Wrote Python scripts to parse JSON documents and load the data in database. Built complete Conguration Management for the Microservices using Kubernetes, Docker and Ansible. Designed front end using HTML, AngularJS, CSS, and JavaScript.
- Implemented SOAP/RESTful web services in JSON format.
- Used PyQT to implement GUI for the user to create, modify and view reports based on client data.
- Worked on Jenkins continuous integration tool for deployment of project. Used IBM DB2 SQL stored procedures and UNIX Shell Scripts for importing/exporting Data and Conversions.
- Created RESTful API's calls with server, parse output report of excel les.
- Designed and managed API system deployment using fast http server and Amazon AWS architecture. Troubleshooting issues by identifying the root cause and nding solutions for the issue. Optimized SQL queries used in batch processing and Wrote validation scripts in SQL to validate data loading.

Environment: Python, Django 1.4, Ruby, HTML5, CSS, XML, MySQL, JavaScript, Eclipse, Git, GitHub, AWS, Linux, Shell Scripting, JIRA, Jenkins, Docker, AngularJS.

Education:

Gannon University, Erie, PA,

Master's in computers and information security |

Aug 2022 – May 2024

Relevant Coursework: Fundamentals of Data Science, Statistical Computing, Advance Statistics, Data Analytics, Security Leadership, Penetration Testing, Asset Security and Risk Management.

Nalla Malla Reddy Engineering College, Hyderabad, India

Bachelor of Technology in Mechanical Engineering |

Projects:

Master's Project: AI-Based Real-Time Email Phishing Detection System

Leveraging Machine Learning to Enhance Cybersecurity by Detecting Phishing Attacks

Problem Solving & System Design: Designed and implemented a comprehensive, real-time email phishing detection system using advanced machine learning (ML) algorithms to detect phishing attacks and safeguard both individuals and organizations.

Data Collection & Preprocessing: Led the data acquisition, cleaning, and preprocessing efforts, extracting valuable features from email headers and body text. Applied feature engineering techniques to transform raw data into actionable insights for training ML models.

Machine Learning & Model Evaluation:

- **Conducted an in-depth evaluation of five ML models:** CatBoost, XGBoost, Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM).
- Performed rigorous model testing across two separate datasets to ensure robustness and generalization of the system.
- Achieved high accuracy and precision in distinguishing phishing from legitimate emails, with Logistic Regression demonstrating exceptional classification performance and CatBoost and Random Forest models excelling in safe email detection.
- **Real-Time Deployment & Testing:** Implemented the developed models in a real-time email phishing detection system and evaluated the system's performance on incoming emails to ensure operational efficiency and security.

Key Skills & Technologies:

- **Machine Learning:** Model selection, training, evaluation, and optimization (CatBoost, XGBoost, LR, RF, SVM).
- **Data Engineering:** Data extraction, transformation, and preprocessing techniques using Python libraries (Pandas, NumPy).
- **Feature Engineering:** Email header and text feature extraction, vectorization (TF-IDF, Word2Vec).
- **Deployment:** Real-time system testing and evaluation.

- **Tools/Technologies:** Python, Scikit-learn, TensorFlow/PyTorch, Jupyter Notebooks, Data Collection & Preprocessing Libraries